

Shadow

A Cryptographically Verifiable Evidence and Spatial Audit
System for AI-Assisted Decisions

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A Cryptographically Verifiable Evidence and Spatial Audit System for AI-Assisted Decisions

From Multi-Agent Answers to Independently Verifiable AI Decision Evidence

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in Computer Science I — practice presentation

The problem

An **AI answer** is not the same thing as **verifiable evidence**.

The gap, concretely:

- ▶ **Source provenance** — what did it actually rely on?
- ▶ **Tool / model actions** — what did it actually do?
- ▶ **Tamper detection** — was the record changed after the fact?
- ▶ **Independent verification** — can *someone else* check it, offline?

Ordinary logs are platform-specific and mutable. A model's explanation is generated prose, not a record.

Project evolution

Orallexa → multi-agent decision support → **Shadow** → portable, verifiable evidence.

- ▶ Orallexa: several AI voices debated an answer. It worked.
- ▶ The lesson: *more opinions is not more trust*.
- ▶ Shadow keeps the analysis as a domain capability — but trust rests on the **evidence layer beneath it**, not on the voices above it.

A refinement of the research question, not a failure of the first one.

What Shadow records

The evidence lifecycle:

Source → Agent / Tool Action → Evidence Event → Hash Chain → Signature → Independent Verification

- ▶ Every event is structured and canonically serialized.
- ▶ Each event commits to the previous one (hash chain).
- ▶ The sealed batch root is signed (Ed25519).
- ▶ Anyone can verify the record — including offline.

Architecture

Core — canonical evidence schema · signed hash-chain record · source maps · verifier · claim–evidence graph

Profiles — banking-v1 · data-science-v1 · coding-agent-v1

Interfaces — CLI · MCP (11 tools) · HTTP · browser
verify.html · Unity Shadow Lens · Three.js replay

Interfaces are of different maturity — the deck says which is which.

Three profiles, one grammar

Profile	Source → action → check
Banking	document → risk claim → source
Data Science	dataset → model → metric
Coding Agent	issue → diff → tests → commit

Same verification grammar underneath: sequence, hash chain, signature, source resolution.

Tamper demonstration the key moment

Pristine → change one earlier event → **first failure** → downstream impact.

- ▶ The verifier reports the **exact first failed sequence** — the earliest event whose hash no longer matches.
- ▶ **Everything after it is invalidated** — the signature covered a chain that no longer exists.
- ▶ Integrity, signature, and *correctness* stay **separate** statuses, even in failure.

This is the strongest thing to show live.

Verify the Verifier

Can you trust the verifier page itself?

- ▶ It carries a **signed fixture manifest**: ASSETS MATCH SIGNED MANIFEST.
- ▶ Independent trust needs an **out-of-band** check: asset hashes vs. a manifest from a separate channel, and the release-key **fingerprint** compared independently — reported as INDEPENDENT COMPARISON NOT PERFORMED.
- ▶ **A page hashing itself is not trust** — and Shadow says so, on the page.

Current signing is fixture, not production.

The spatial experience

Unity **Shadow Lens** + Three.js replay — 3D is for **sequence, provenance, and tamper replay**.

- ▶ Three workspaces; provenance audit arc; Verify / Tamper / Reset.
- ▶ **Head-directed focus** (gaze interactor) — hover only. *Not eye tracking. No RGB. No 6DoF on the mock.*
- ▶ Android **mock APK is built** (24.4 MB, hashed). **Beam Pro / XREAL device validation is pending.**
- ▶ A precise **2D fallback** always remains.

Evaluation

- ▶ **1,824 / 1,827 host tests pass** (0 fail, 3 env-gated skips), re-run 2026-07-21.
- ▶ **Browser acceptance:** Chromium 149 (Playwright), EN + , 8 flows, CSP **0 external / 0 violations**, offline verify — real screenshots.
- ▶ **Android:** mock APK **built** (24,442,084 B, SHA-256 93f2a81a...). *Built, not device-validated.*
- ▶ Three.js: 4 layouts rendered + recorded. **No user study yet.**

Where it stands

Implemented now	Pending next (Capstone II)
Deterministic evidence + hash chain	Beam Pro / XREAL device validation
Independent + offline verification	Shared Unity/Three.js scene contract
Exact tamper localization	User study (does 3D actually help?)
Bilingual verifier (EN / zh-CN)	Semantic audit of ingested output
Android mock build	Production signing · KMS/HSM

Fixture-signed · device-pending · no correctness claim — stated honestly.

Contribution

Shadow does **not** ask the auditor to trust the AI's answer.

It gives the auditor evidence that can be independently verified — the sequence, the sources, the signature, and, if it was tampered with, the exact point where the record breaks.

And it draws a firm line: **integrity, not correctness.**

Thank you — questions welcome.